Multi-Regime Red Teaming of Machine Unlearning

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Abstract

Machine unlearning methods are supposed to make models forget specific training data, but in practice they often retain that information in subtle ways. I study how recoverable this hidden knowledge is across eight unlearning methods—SalUn, SCRUB, RURK, CE-U, SGA, BalDRO, ORBIT, and robust-base SalUn—attacking them four ways: universal perturbations, targeted per-example conditional recovery, local patch-based attacks (single patch, border frame, multi-patch), and amortized short-budget attacks on a controlled CIFAR-10 benchmark. The main finding is that the threat is not one-size-fits-all: different methods fail under different attacks, and a defense that works against one type often does not transfer to another. Concretely, patch-aware stage-2 defenses reduce 32×32 conditional patch recovery on ORBIT (78.1%→31.2%) and CE-U (68.8%→43.8%) but fail against border-frame and multi-patch delivery, establishing local-delivery transfer as an open problem. I also show that a post-hoc feature-subspace defense partially reduces recoverability on BalDRO, SalUn, and ORBIT without collapsing clean accuracy. The results suggest that evaluating unlearning on clean forgetting accuracy alone is insufficient—the residual knowledge is there, it is just hard to see. Code and checkpoints are available at https://github.com/DevT02/ForgetGate.

1 Introduction

Machine unlearning methods are supposed to make models forget specific training data. Most papers check this by looking at clean forget accuracy—does the model still predict the forgotten class? I looked at this differently: what if you actively try to recover the forgotten information using adversarial attacks?

I evaluated eight unlearning methods under four attack families—per-example conditional recovery, universal perturbations, conditional patches, and border-frame and multi-patch variants—on a controlled CIFAR-10 benchmark. The short version is that no single attack breaks everything, but no single method is safe either. Different methods fail in different ways. Some are vulnerable to targeted per-example recovery but not patches. Some look robust until you try a slightly different attack surface. And a defense that closes one hole often leaves another open.

That framing raises a harder question than simple attack success: when a perturbation recovers the forgotten class, is it actually exposing residual class-specific information that the unlearning method failed to erase, or is it merely inducing a generic adversarial target-label failure? The paper is designed to separate those cases. I therefore compare forget-set examples against matched retain-class controls under the same targeted recovery audit. If forget examples are selectively easier

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to push back to the forgotten class, that supports a residual-knowledge interpretation. If forget and retain controls are equally easy to hijack, the result is better understood as generic adversarial fragility than as genuine recovery of supposedly forgotten information.

The positive results are modest but real: a post-hoc feature-subspace defense consistently reduces recoverability on several methods without hurting clean accuracy, and patch-aware training partially closes the patch channel on ORBIT and CE-U—though it does not transfer to frame or multi-patch delivery. The benchmark, code, and checkpoints are fully reproducible at `https://github.com/DevT02/ForgetGate`.

2 Threat Models and Attack Regimes

2.1 Universal perturbations

The first regime includes Fourier, tile, warp, and codebook-style universal perturbations. These attacks are useful primarily as a negative result: on stronger methods, the residual vulnerability is not well-compressed into one shared low-complexity direction.

2.2 Per-example conditional recovery

The main broad attack optimizes the forgotten-class target margin

$$m_t(x) = z_t(x) - \max_{j \neq t} z_j(x), \tag{1}$$

using strong first-order optimizers such as `adam_margin` and `apgd_margin`. This regime asks whether a forgotten behavior is locally recoverable by a targeted per-example perturbation inside a fixed norm budget.

2.3 Amortized short-budget attacks

I additionally study a learned image-conditional perturbation generator trained from short Adam teacher trajectories and refined with a tiny local budget. This attack family is not a universal win, but it improves matched low-step attacks on ORBIT and SalUn.

2.4 Conditional patches

I also study a conditional small-area patch regime in which both patch content and placement depend on the input. This is a patch-based regime, not a universal trigger claim.

2.5 Additional local delivery surfaces

To test whether patch-aware defenses generalize beyond one square patch, I also study conditional border-frame attacks and conditional multi-patch attacks. These are stronger local-delivery checks than a single-patch evaluation because they alter geometry while preserving the same selective red-team objective.

3 Related Work

These experiments sit at the intersection of machine unlearning, adversarial robustness, and parameter-efficient adaptation. I build directly on SalUn [2], SCRUB [1], LoRA [3], VPT [5], and

Method / Row	Fidelity	Paper Role	Current Read
SalUn (faithful full-ft)	faithful	main text	core literature anchor
SCRUB distill	reasonably faithful	main text	literature baseline; hard negative / generic fragility
ORBIT	repo-defined	main text	internal benchmark row; best later defense row
RURK	reasonably faithful	main text	recent method; hard negative / resistant case
SGA	adapted recent method	auxiliary	generic-fragility cross-domain benchmark row
BalDRO	adapted recent method	main text	main cross-domain benchmark row
CE-U	reasonably faithful	main text	recent related-work addition
robust-base SalUn	faithful frontier	main text	utility-vs-recovery frontier
Uniform KL	bridge / supporting	appendix	seed-sensitive bridge target
Prune+KL	supporting	appendix	auxiliary clean-forgetting row
CNN cross-checks	supporting	appendix	architecture sanity checks
legacy manifold/imposter	supporting	appendix	older harder-than-baseline branch
LoRA SalUn	proxy	appendix	PEFT-side approximation only
FaLW variants	proxy or invalid	appendix	not usable under current benchmark
Langevin variants	proxy or invalid	appendix	not valid forgetting rows yet
OrthoReg / noisy_retain	proxy	appendix	repo-side approximations only

Table 1: Method classification used in the paper. The table separates literature anchors, appendix-only supporting rows, and proxy artifacts so that the main benchmark reflects only rows with defensible evaluation fidelity.

AutoAttack [4], and I treat recent 2025–2026 methods such as RURK [6] and CE-U [11] as newer literature baselines. I additionally include adapted recent objectives such as SGA [8] and BalDRO [9] as cross-domain benchmark rows, and one repo-defined internal row, ORBIT, as a useful stress-test target rather than as an external literature anchor. I also keep several additional repository rows, including FaLW [12] and Langevin-style [10] variants, only as proxy or supporting comparisons when they help define regime boundaries. I use Langevin Unlearning [10] as a theoretical anchor for principled privacy-certified evaluation and MUSE [7] as an evaluation-oriented benchmark perspective to motivate the paper's emphasis on multi-regime testing rather than single-metric forgetting claims.

4 Experimental Setup

4.1 Models and data

All experiments use CIFAR-10 with 224×224 inputs and ViT-tiny-based checkpoints. The main benchmark rows in the paper are SalUn and SCRUB as literature anchors, RURK and CE-U as newer literature methods, SGA and BalDRO as adapted recent methods, and ORBIT as a repo-defined internal benchmark row, together with robust-base SalUn as a defense frontier. I also retain several appendix-only proxy, bridge, and negative-control rows already present in the repository.

4.2 ORBIT: repo-defined benchmark method

ORBIT (oracle-aligned, representation-dispersive unlearning) is a method defined in this repository, used as a stress-test target rather than as an external literature contribution. Its objective combines four terms: (1) a hinge-margin loss suppressing the forget-class logit below the maximum non-forget logit; (2) masked dark-knowledge distillation from a frozen oracle on non-forget classes, with the forget dimension excluded from the KL target; (3) a feature alignment term pulling forget-input representations toward the oracle's feature space; and (4) a dispersion penalty pushing forget features apart to weaken prototype and linear recovery channels. The motivation is that suppressing a

Regime	Threat Model	Primary Metric	What Counts as a Defense Win
Conditional recovery	per-example full-image targeted recovery	forget median / retain median / clean-success	higher forget median without comparable retain-side degradation
Conditional patch	local one-patch delivery	forget success / retain-to-forget success / alpha	lower forget success without added retain-to-forget failure
Conditional frame	border-frame local delivery	forget success / retain-to-forget success / alpha	lower forget success without transfer failure on retain
Conditional multi-patch	multi-local delivery	forget success / retain-to-forget success / alpha	lower forget success under richer local geometry
Amortized short-budget	learned low-step attack	forget success / retain success / median radius	stronger selective low-step recovery than plain low-step optimizer
Robust-base frontier	utility-recovery tradeoff	retain accuracy vs recovery radius	better operating point, not a binary robust point

Table 2: Protocol table for the benchmark. The benchmark is designed as a controlled robustness microscope: each regime tests a distinct failure mode, and a defense is only treated as positive when it improves the regime-relevant robustness metric without introducing comparable retain-side failure.

single output logit is insufficient when the underlying representation still encodes the forgotten concept—alignment and dispersion together target the representational substrate rather than only the output layer. ORBIT is the strongest patch-positive case in the benchmark, making it the primary vehicle for studying local-delivery attacks and patch-aware defenses.

4.3 LoRA as benchmark substrate

The repository originated as a LoRA-centered audit, and many of the stored checkpoints remain LoRA adapters for comparability and fast stage-2 iteration. That implementation choice is useful experimentally, but it is not the main scientific claim of the paper. The paper is about unlearning robustness under multiple attack regimes. I therefore treat LoRA primarily as part of the benchmark substrate, while using selected full-finetune and cross-architecture rows to verify that the resulting story is not purely an artifact of one parameter-efficient adaptation scheme.

4.4 Baseline checkpoint quality

The clean baseline summary shows that unlearning methods can appear successful on clean metrics while still exposing recovery channels. Across seeds 42, 123, and 456, clean retain accuracy stays in the low 90s for the main baselines, while forget accuracy varies dramatically by method. The base model sits at 91.83 ± 4.36 forget and 91.80 ± 3.77 retain. Uniform KL, CE Ascent, SalUn, and SCRUB all preserve similar retain performance, but their forget-side variance is extremely large, which already suggests that clean-point forgetting is an unstable proxy for actual robustness. These clean baselines therefore motivate attack-based evaluation rather than one-number forgetting claims.

4.5 Recovery metric

For each forget or retain-control sample, I search for the minimum perturbation that causes the unlearned model to predict the forgotten class. I report the median successful radius and clean-success rate under a matched audit battery. The interpretive point is not attack success in isolation. The key question is whether forget examples are selectively easier to recover than matched retain controls. A consistent forget-vs-retain gap supports the view that the attack is exposing residual information left behind by incomplete unlearning, whereas non-selective success on both groups is more consistent with generic adversarial fragility.

5 Multi-Regime Results

I evaluate machine-unlearning robustness under four complementary attack regimes: strong per-example conditional recovery, conditional patches, border-frame delivery, and multi-patch local

Model	Setup	Forget Median	Retain Median	Clean Success
BalDRO	baseline	0.00343	0.00846	0.000
BalDRO	feature-subspace stage-2	0.00349	0.00858	0.000
SalUn	baseline	0.00251	0.00564	0.031
SalUn	feature-subspace stage-2	0.00276	0.00600	0.031
ORBIT	baseline	0.00340	0.00643	0.031
ORBIT	feature-subspace stage-2	0.00423	0.00690	0.000
RURK	baseline	0.00159	0.00576	0.062
RURK	feature-subspace stage-2	0.00159	0.00576	0.062
CE-U	baseline[†]	0.00368	—[‡]	0.000
SCRUB	baseline only	0.01078	0.00980	0.000

[†]CE-U: 2-step pilot audit, n=8, seed 42.

[‡]0 of 8 retain-control attacks succeeded within ϵ; retain median undefined.

Table 3: Recovery-radius medians under the strongest per-example conditional attack. Feature-subspace stage-2 helps BalDRO, SalUn, and ORBIT; leaves RURK and CE-U unchanged; SCRUB is a negative control with nearly identical forget and retain radii. CE-U retains 0 of 8 retain-control attacks in the pilot audit, indicating stronger selectivity than methods with high retain-side attack success.

delivery. Across the main benchmark rows audited in depth—SalUn, SCRUB, RURK, CE-U, BalDRO, and the repo-defined ORBIT row—the dominant threat remains targeted per-example recovery, while universal perturbations are mostly ineffective on the stronger checkpoints. The point of this matched audit is not to ask whether a target-class adversarial perturbation exists in the abstract. It is to ask whether forget examples are selectively easier to drive back to the forgotten class than matched retain controls. Figure 1 and Table 3 summarize the main defense result: a feature-subspace stage-2 defense modestly increases recovery radii on BalDRO, SalUn, and the repo-defined ORBIT row without collapsing clean accuracy. Figure 2 shows that robust-base SalUn does not eliminate leakage outright; instead it moves along a utility-vs-recovery frontier. Figure 3 shows that conditional patches remain viable, but only for a subset of methods.

Feature-subspace stage-2 defense. The first defense family that consistently helps is a post-unlearning feature-subspace stage-2 finetune. Rather than modifying the entire forgetting objective from epoch 0, I begin from an already-unlearned adapter and suppress attacked forget-sample features inside a learned recovery subspace. This regime avoids the instability of earlier logit-margin defenses and produces small but repeatable gains on BalDRO, SalUn, and the repo-defined ORBIT row. The effect is modest in absolute magnitude because the benchmark is CIFAR-10 under an L_∞ budget on 224×224 inputs, where successful radii are numerically small by construction. What matters is that the defense shifts the forget-vs-retain frontier in the right direction without sacrificing clean retain utility.

CE-U as a core baseline. CE-U belongs in the main benchmark even though its first qualified positive defense row appears in the patch regime rather than under full-image recovery. Under the 2-step pilot audit (n=8, seed 42), the baseline CE-U checkpoint has forget median 0.00368 with 0% clean forget success, and retain-control attacks succeed 0 of 8 times within the ϵ budget—stronger selectivity than SalUn or ORBIT, where retain attacks succeed at high rates under full audits. CE-U full-image feature-subspace stage-2 variants are at best flat, and the combined row is slightly worse under the APGD cross-check, so the CE-U defense claim is reserved for the qualified combined

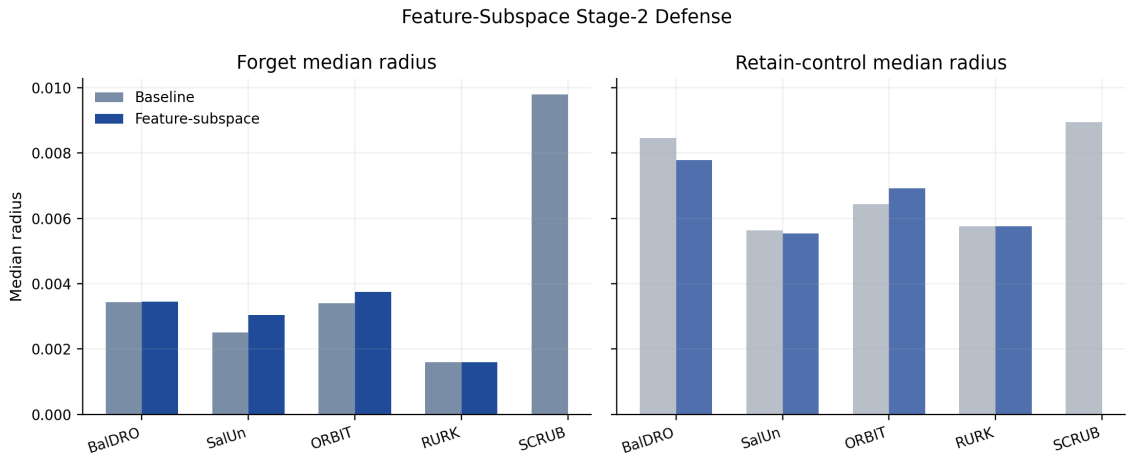

Figure 1: Feature-subspace stage-2 tradeoff. Larger forget medians indicate stronger resistance to conditional recovery. The defense yields small but consistent improvements on BalDRO, SalUn, and the repo-defined ORBIT row, while RURK remains hard.

stage-2 patch result discussed below.

Robust-base frontier. Robust-base SalUn does not behave like a binary defense success. Instead, it exposes a frontier: at low utility, conditional recovery can drop to zero on the sampled audit; as retain accuracy rises, recovery reappears. This is a stronger framing than a single checkpoint claim because it shows that robustness and post-unlearning utility are coupled. I therefore treat robust-base training as an operating-point choice rather than a one-shot fix.

Conditional patches. The patch result is not a universal trigger. Universal static patches were mostly ineffective on the stronger methods. The viable regime is instead a conditional small-area patch attack whose content and placement depend on the input. The repo-defined ORBIT row is the flagship positive case: a 32×32 patch covering 2.04% of the image recovers forgotten behavior strongly while inducing no retain-to-forget hijack. SalUn is patch-vulnerable only at a larger 48×48 budget, and BalDRO is effectively negative at both tested patch sizes. This method dependence reinforces the broader paper thesis: residual forgetting leakage is structured, but not universal.

Delivery surfaces beyond one patch. The later delivery-surface checks strengthen the attack story and narrow the defense story. Conditional frame attacks and conditional multi-patch attacks are both real local delivery surfaces, but they do not behave like simple extensions of the one-patch regime. Patch-aware defenses that help on one square patch often fail to transfer: the ORBIT patch-stage2 row becomes easier to recover under the frame attack, CE-U combined stage-2 also becomes easier and adds retain-side failure under the frame attack, and faithful full-ft SalUn patch-stage2 becomes easier under multi-patch delivery. The main exception is weak positive transfer from the ORBIT patch-stage2 row to multi-patch delivery. I therefore treat frame and multi-patch regimes as first-class red-team surfaces and not as appendix-only perturbation variants.

Second-wave defenses. Two later branches sharpen the story. First, teacher-guided stochastic CVaR provides a small but reproducible full-image defense improvement on the repo-defined ORBIT row across two seeds, whereas the same probabilistic family remains neutral on CE-U. Second, a

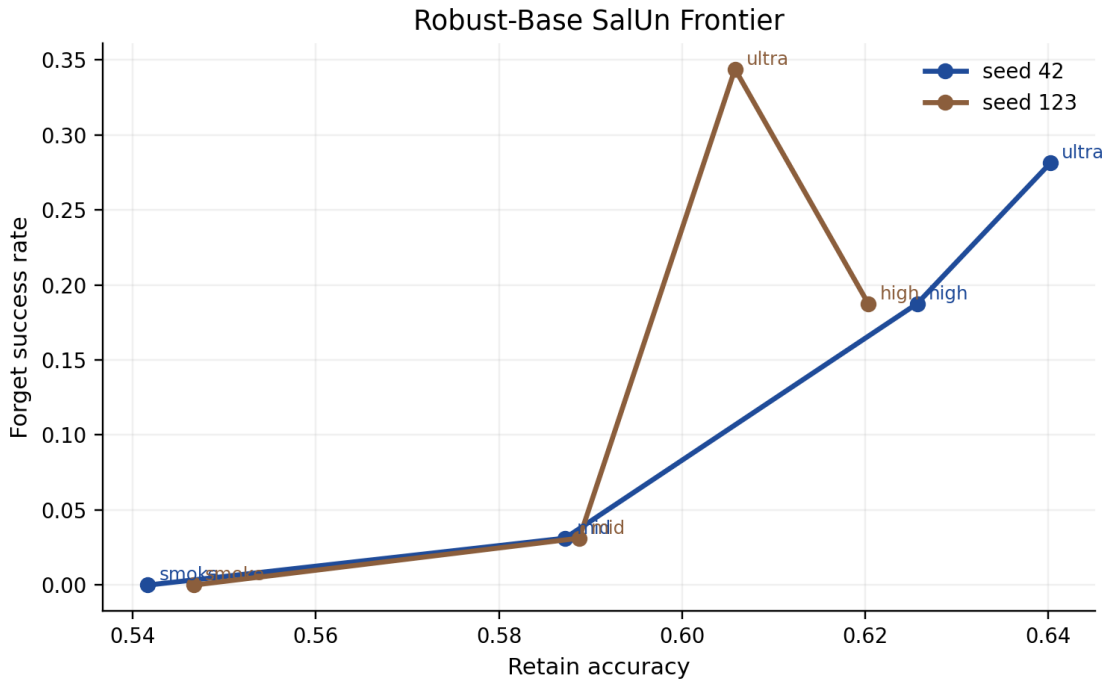

Figure 2: Robust-base SalUn frontier. Lower-utility robust-base checkpoints suppress conditional recovery more strongly, but leakage returns as retain utility improves.

combined subspace-plus-patch stage-2 defense gives the first CE-U row that improves under matched Adam audits while also reducing conditional patch success, although its APGD result is still slightly worse than baseline. By contrast, the larger mixed-surface stage-2 branch was explicitly designed to transfer across one-patch, frame, and multi-patch delivery and failed to do so reliably. These later rows are therefore useful but qualified: they extend the defense story beyond pure feature-subspace training without becoming universal fixes.

Takeaway. The combined evidence supports a multi-regime view of red-teaming machine unlearning. Strong per-example conditional attacks remain the main threat, conditional patches expose an additional local attack surface on some methods, and frame plus multi-patch attacks show that local-delivery robustness is materially harder than one-patch robustness alone. The best defense families I found are modest and method-specific: feature-subspace stage-2 helps BalDRO, SalUn, and the repo-defined ORBIT row; teacher-guided stochastic CVaR helps the repo-defined ORBIT row slightly; and CE-U combined stage-2 is only a qualified partial row. RURK remains a hard case, and SCRUB serves as a negative control where generic adversarial fragility overwhelms forget-specific interpretation.

6 Short-Budget and Auxiliary Regimes

6.1 Amortized short-budget attack

The learned distilled short-budget attacker is a real but target-dependent contribution. It improves matched low-step Adam on the repo-defined ORBIT row across both seeds and also helps on

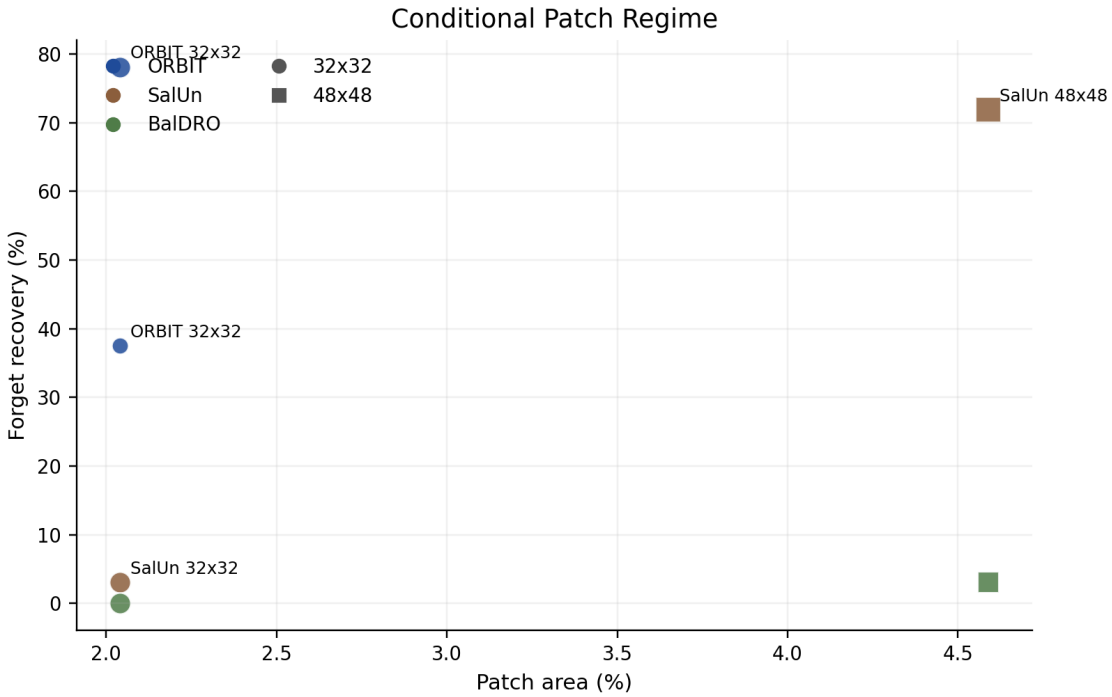

Figure 3: Conditional patch regime. The repo-defined ORBIT row admits strong low-area patch recovery, SalUn requires a larger patch and incurs more retain damage, and BalDRO is effectively negative under the same patch family.

SalUn, but it does not preserve a clean forget/retain Pareto on RURK. I therefore position it as an amortized attack regime rather than a universal attack method.

6.2 Earlier prompt/probe regime

An earlier prompt-style red-team branch found a strong effect on Uniform KL under an oracle-contrastive probe, but not on the stronger methods. In particular, the strongest earlier prompt result moved Uniform KL seed 42 from 12.70% forget to 21.10% forget with only a 0.36pp retain drop, while SalUn, SCRUB, and the repo-defined ORBIT row remained in the resistant bucket under that family. I treat this as supporting evidence for regime sensitivity rather than as the core attack result.

6.3 Uniform KL as a bridge target

Uniform KL also remains useful under the newer conditional-recovery regime. A fast matched pilot with 2-step `adam_margin` on seed 42 recovers all sampled forget examples with median radius 0.00147 while succeeding on only one of eight sampled retain controls. The same pilot on seed 123 is completely flat. This makes Uniform KL a bridge target between the older probe story and the newer conditional-optimization story: it is not only probe-vulnerable, but also locally recoverable on the vulnerable seed. A conditional 32×32 patch on seed 42 also yields a weak but nontrivial patch result (15.6% forget recovery and 0% retain-to-forget success), which is enough to show that KL is patch-accessible without making it a flagship patch case.

Target	Seed	Attack	Forget Succ.	Retain Succ.	Forget Median
ORBIT	42	2-step Adam	9.4%	0.0%	0.02500
ORBIT	42	Distilled + 2-step refine	18.8%	0.0%	0.02083
ORBIT	123	2-step Adam	18.8%	0.0%	0.00316
ORBIT	123	Distilled + 2-step refine	31.2%	3.1%	0.01189
SalUn	42	2-step Adam	71.9%	9.4%	0.00435
SalUn	42	Distilled + 2-step refine	84.4%	9.4%	0.00447
SalUn	123	2-step Adam	68.8%	6.2%	0.00328
SalUn	123	Calibrated distilled + 2-step refine	81.2%	12.5%	0.00389
RURK	42	2-step Adam	87.5%	6.2%	0.00175
RURK	42	Calibrated distilled + 2-step refine	90.6%	18.8%	0.00175

Table 4: Short-budget comparison between matched 2-step Adam and the learned distilled conditional attacker. The learned attacker helps most on ORBIT, helps on SalUn with a weaker forget/retain Pareto, and does not remain selective on RURK.

Target	Seed	Regime	Forget Succ.	Retain Succ.	Forget Median / Alpha
Uniform KL	42	2-step conditional pilot	100.0%	12.5%	0.00147
Uniform KL	123	2-step conditional pilot	0.0%	0.0%	N/A
Uniform KL	42	32×32 conditional patch	15.6%	0.0%	0.15234
Uniform KL	42	feature-subspace stage-2 pilot	100.0%	12.5%	0.00147
SGA	42	full conditional audit	100.0%	100.0%	0.00294
SGA	123	full conditional audit	96.9%	100.0%	0.00968
SGA	456	full conditional audit	96.9%	96.9%	0.01091
SGA	42	32×32 conditional patch	0.0%	0.0%	N/A
SGA	42	2-step baseline pilot	75.0%	0.0%	0.00417
SGA	42	2-step stage-2 pilot	62.5%	0.0%	0.00441

Table 5: Auxiliary seed-sensitive targets. Uniform KL is a bridge case between the older probe regime and the newer conditional-recovery regime, while SGA behaves like generic adversarial fragility under the full audit, stays patch-negative, and only shows a weak fast-pilot stage-2 effect.

6.4 Additional auxiliary targets

SGA is another useful auxiliary target from the recent related-work list, but the full conditional audit does not support a selective-leakage reading. Across seeds 42, 123, and 456, forget and retain-control success rates are both near ceiling, so this row looks like generic adversarial fragility rather than a meaningful forget-vs-retain gap. A matched 32×32 conditional patch is negative on seed 42, and a lightweight feature-subspace stage-2 pilot yields only marginal improvement. I therefore treat both Uniform KL and SGA as auxiliary regime-defining targets rather than as the main methods around which to build the defense story.

7 Defense Discussion

7.1 Feature-subspace stage-2

The first defense family that consistently helps is a post-unlearning feature-subspace stage-2 finetune. Instead of modifying the full unlearning objective from epoch 0, I begin from an already-unlearned adapter and suppress attacked forget-sample features inside a learned recovery subspace. This avoids the instability of earlier logit-margin and teacher-distillation defenses on BalDRO.

7.2 Why the gains are small

The absolute changes in reported radii are numerically small because the benchmark itself is small: CIFAR-10 under an L_∞ budget on 224×224 inputs yields radii on the order of a few thousandths. The right question is therefore relative: does the defense move the forget-vs-retain frontier without collapsing clean utility? On BalDRO, SalUn, and the repo-defined ORBIT row, the answer is yes, though modestly.

7.3 What does not generalize

The defense is not universal. RURK remains a hard case under both attack and defense. SCRUB behaves like generic adversarial fragility rather than a clean forgetting-leak regime, so stronger attacks on SCRUB do not necessarily imply a meaningful unlearning failure mode. The newer delivery-surface results make this sharper: patch-aware defenses often fail to transfer to border-frame or multi-patch delivery, and the mixed-surface stage-2 branch does not reliably fix that.

8 Proposed Defense Objective

The experiments suggest that clean-point forgetting is too weak a training target. A more principled direction is targeted-margin adversarial unlearning:

$$\min_{\theta} \mathbb{E}_{x\in R} L_{\text{retain}}(x;\theta) + \lambda_f \mathbb{E}_{x\in F} L_{\text{forget}}(x;\theta) + \lambda_m \mathbb{E}_{x\in F} \max(0,\ m_t(x+\delta^*(x)) + \gamma)\,, \tag{2}$$

where F is the forget set, R is the retain set, m_t is the forgotten-class target margin, $\delta^*(x)$ is an inner-loop or amortized attack, and γ is a safety margin. The implemented feature-subspace stage-2 defense can be viewed as a computationally cheaper approximation to this idea: defend against locally recoverable targeted behavior, not just against clean forget accuracy.

9 Discussion

The strongest framing of this work is a multi-regime red-team study rather than a single-method attack paper. The evidence does not support a universal "magic" attack or defense. Strong per-example conditional recovery is the main general threat; universal perturbations are mostly weak on the stronger checkpoints; conditional patches expose an additional local attack surface on some methods; border-frame and multi-patch delivery show that local-delivery transfer is a separate problem; robust-base methods define a utility-vs-recovery frontier rather than a binary robust point; and feature-subspace stage-2 provides the first modest defense gains on several targets. Later defense branches remain method-specific: teacher-guided stochastic CVaR helps the repo-defined ORBIT row slightly but not CE-U, while patch-aware stage-2 gives the first clearly positive CE-U result in the patch regime (68.8%→43.8% at 32×32, seed 42) with no retain-side failure, though it does not help on the full-image conditional attack. Mixed-surface stage-2 is informative precisely because it fails: even explicit delivery-surface-aware training does not reliably generalize across one-patch, frame, and multi-patch attacks. Uniform KL reinforces this framing: seed 42 is vulnerable under both the older prompt/probe family and the newer conditional audit, while seed 123 stays flat under the same fast conditional attack. That multifaceted pattern is scientifically stronger than forcing all methods into one attack or defense family.

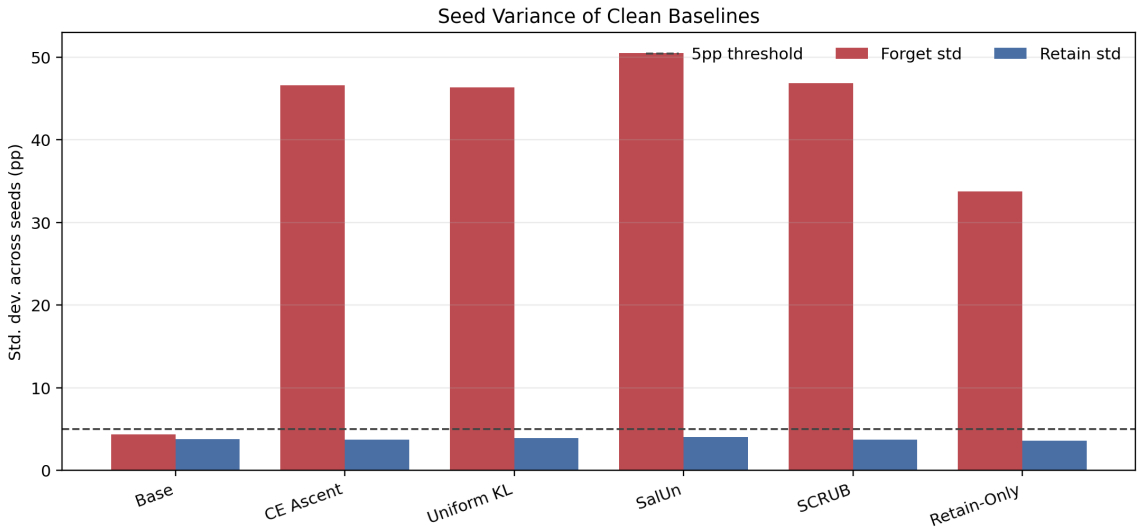

Figure 4: Seed variance of clean baselines. Forget-side variance is large for several methods even when retain-side variance remains small.

10 Limitations

I do not claim a universally stronger new optimizer than strong first-order attacks such as `adam_margin` or `apgd_margin`. The amortized short-budget attack is partial and target-dependent. The defense gains are modest and currently demonstrated on a benchmark that is intentionally controlled rather than broad: mostly CIFAR-10 ViT checkpoints, often with LoRA-based unlearning implementations, plus a smaller number of full-finetune and cross-architecture checks. This makes the benchmark useful for discovering failure modes and transfer gaps, but not yet a direct proxy for most industry deletion settings. Finally, some regimes are better understood as negatives or controls than as headline wins.

11 Conclusion

Machine unlearning cannot be evaluated reliably with clean forget accuracy alone. Across this benchmark, the dominant leakage channel is strong per-example conditional recovery, but local patch, frame, and multi-patch attacks expose additional method-specific surfaces that do not reduce to a single universal vulnerability. The defense picture is similarly non-universal: feature-subspace stage-2 helps several rows, patch-aware training helps ORBIT and CE-U locally, and transfer across delivery surfaces remains unsolved.

A Appendix: Seed Variance and Baseline Stability

The clean baseline regime itself is highly variable across seeds for several unlearning methods, especially on forget accuracy. Figure 4 summarizes this instability. This is one reason the paper emphasizes attack-conditioned evaluation rather than clean forget accuracy alone.

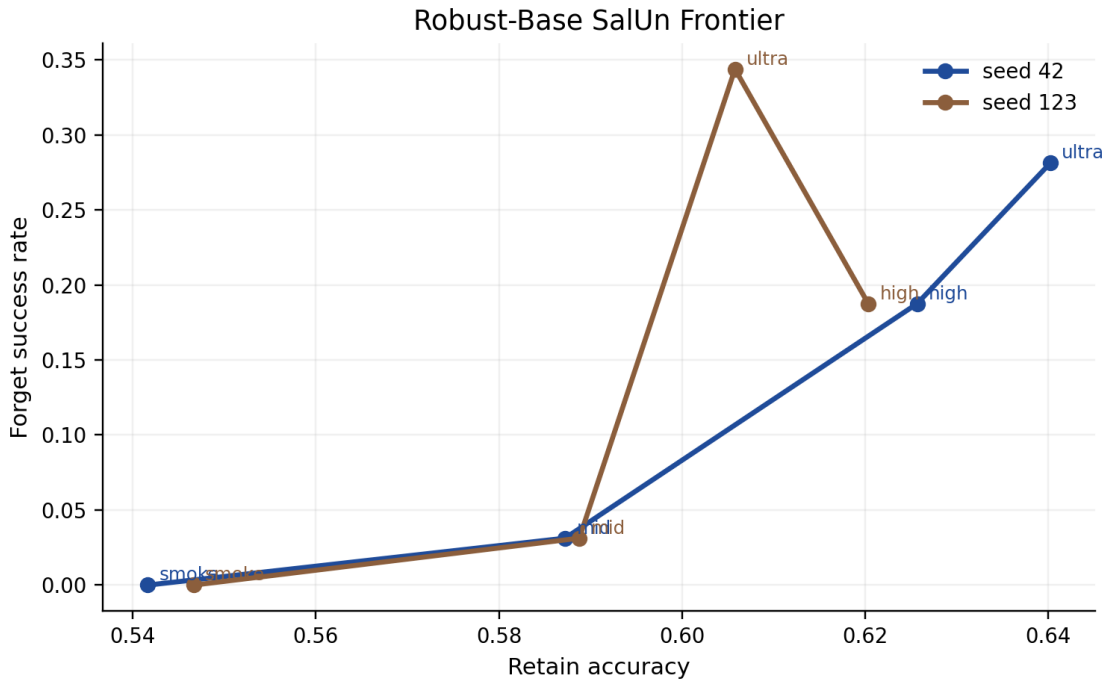

Figure 5: Robust-base SalUn operating-point frontier. Conditional recovery re-emerges as retain accuracy rises, confirming that robust-base training shifts the frontier rather than eliminating leakage.

B Appendix: Robust-Base Frontier

Figure 2 summarizes the robust-base SalUn operating-point story. At low utility, recovery can be suppressed strongly or disappear on the sampled audit. As retain accuracy increases, conditional recovery re-emerges. This makes robust-base training a frontier choice rather than a single robust point.

C Appendix: Conditional Patch Regime

The patch regime is method-dependent. The repo-defined ORBIT row is the flagship positive case with a 32×32 conditional patch covering 2.04% of the image, strong forget recovery, 0% retain-to-forget success, and limited retain accuracy drop. SalUn is patch-vulnerable only at a larger 48×48 patch budget. BalDRO is effectively negative at the tested budgets. Figure 3 summarizes the patch success rates across methods and seeds.

D Appendix: Delivery-Surface Transfer

The later delivery-surface checks separate one-patch wins from broader local-delivery robustness. Under the conditional frame attack, the ORBIT patch-stage2 row and CE-U combined stage-2 both become easier to recover, and CE-U also incurs retain-side failure. Under conditional multi-patch delivery, the ORBIT patch-stage2 row transfers only weakly, while CE-U combined stage-2 and faithful full-ft SalUn patch-stage2 become easier to recover. Mixed-surface stage-2 was designed

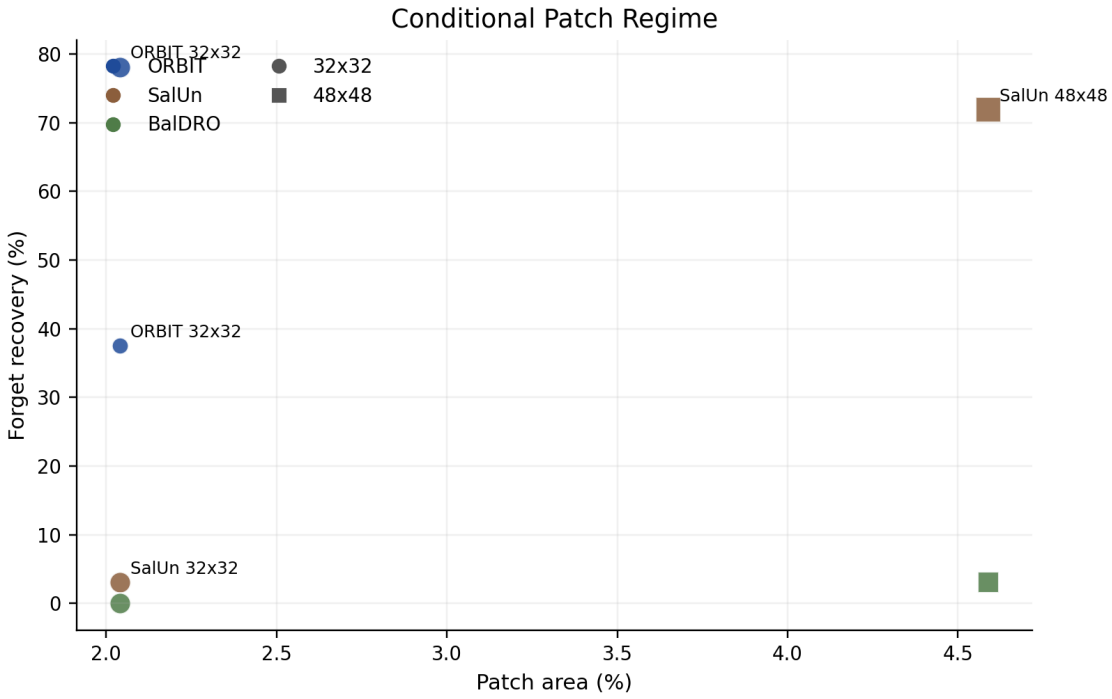

Figure 6: Conditional patch recovery rates across methods. ORBIT is the strongest patch-positive case; BalDRO and robust-base SalUn are effectively negative at 32×32.

to fix this transfer problem, but in practice it only gives narrow gains and often worsens frame or multi-patch robustness. I therefore treat delivery-surface transfer as an open defense problem rather than as solved by the current stage-2 families.

E Appendix: Negative and Hard Cases

RURK remains a hard case for both the amortized attack family and the feature-subspace defense. SCRUB is useful as a negative control: strong conditional attacks succeed, but forget and retain-control radii are almost identical, which suggests generic adversarial fragility rather than a clean forgetting-leak mechanism. Several later defense branches also belong in this bucket: mixed-surface stage-2, frame-transfer failures, and the faithful full-ft SalUn mixed-surface row are all informative mainly because they show what does not transfer.

F Appendix: Reproducibility Artifacts

The paper draws from repository summaries and generated figures in:

- `results/analysis/feature_subspace_tradeoff.md`
- `results/analysis/recovery_tradeoff_summary.md`
- `results/analysis/conditional_patch_summary.md`
- `results/analysis/paper_results_outline.md`

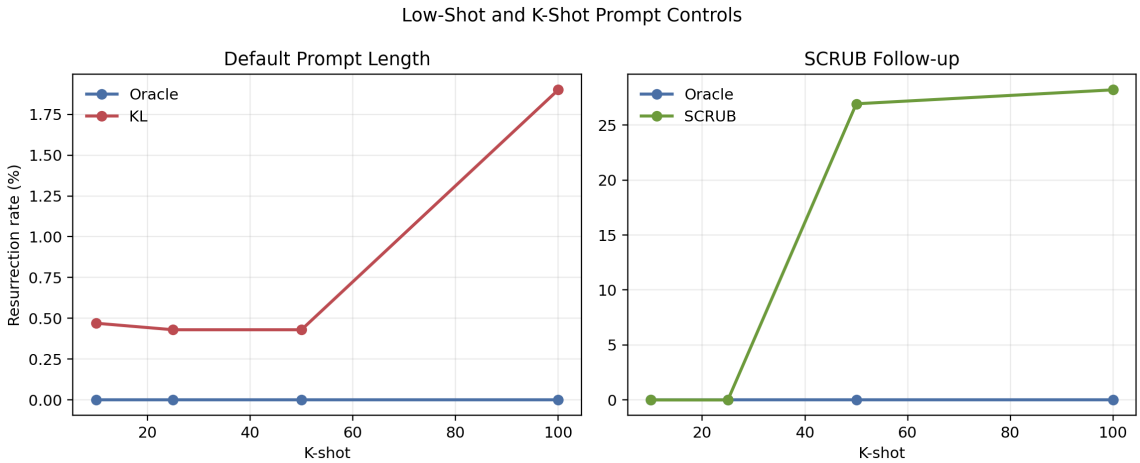

Figure 7: K-shot prompt controls. Oracle stays flat, KL remains weakly vulnerable, and SCRUB only becomes unstable in the larger-shot follow-up.

- `results/analysis/figures/`

All figures and tables in this paper are generated from JSON result files committed alongside the code. The repository contains checkpoints for all eight benchmark methods across seeds 42, 123, and 456 together with the audit scripts needed to reproduce each attack regime.

G Appendix: Low-Shot Controls

The lower-shot prompt controls help separate real recovery from trivial prompt memorization. Figure 7 summarizes two useful trends: oracle controls remain flat near zero across the tracked shot counts, while KL and SCRUB can become vulnerable only in specific low-shot or high-shot regimes.

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